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Integrated Data Mining and Automation for Gas Lift Design Optimization

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Abstract

This paper presents an innovative approach that integrates data mining and automation to enhance gas lift design across the entire lifespan of a well, leveraging the capabilities of Industrial Revolution (IR) 4.0 technologies. The primary goal is to streamline the design process through advanced analytics, ensuring optimal performance in the increasingly digitalized oil and gas sector.

The methodology draws on reservoir data, PVT analysis, well completion details and surface facilities information. By combining data mining with automation tools, the entire process is automated to provide a bespoke gas lift design. A digital twin of the well is created using industry standard production engineering software, integrated with advanced data visualization and analytics platforms. This system efficiently processes data, enabling seamless modelling and optimization utilizing key gas lift design parameters such as reservoir pressure, productivity index (PI), gas liquid ratio, casing heads pressure, injection depth, lift gas gravity, kill fluid gradient and flowing wellhead pressure.

The proposed method has been successfully applied using pre-existing data. Through over 100 simulation scenarios, the approach optimizes valve selection, size and spacing. Advanced analytics guide the selection of multiple valve depth combinations for further evaluation, ensuring the most effective gas lift design over well's life cycle. This approach also identifies optimal times for valve replacement or upgrades. By automating this process, design time is reduced from one day to under an hour, while maintaining a transparent and auditable decision-making framework.

This methodology represents a significant step in the digital transformation of the oil and gas industry, combining data mining, automation and IR 4.0 concepts. By incorporating advanced analytics and automation, this approach not only enhances gas lift design but also improves overall operational efficiency, reduces costs and optimizes production through the well's lifecycle. The work highlights how data driven decision making and automation can lead to more efficient and cost-effective operations in today's digitally driven industry.

Introduction

Production of hydrocarbons is aided by the artificial lift systems. Among the several artificial lift systems including Electric Submersible Pumps (ESP), rod pumps, progressive cavity pumps and plunger lifts-gas lift stays one of the most flexible and adoptable technologies. High deviation, offshore or deep wells where

mechanical pumps face reliability and deployment restrictions are particularly suited for it. Key benefits of gas lift are its capacity to support continuous production with minimum surface infrastructure, tolerate solids and water-cut and manage changing flow conditions (Adiyodi et al. 1999).

Evaluating reservoir inflow, modelling TCA pressure profiles, choosing valve depths, port sizes and determining dome pressures from a multistep process in traditional gas lift design are usually executed using deterministic approaches. These steps demand major human involvement as well as engineering judgement. This system has several drawbacks in the fast-changing digital oil fields of today. Static design often overlooks changing reservoir conditions over the well lifetime. Manual input and data transfer run the risk of inconsistency and delays and scenario-based optimization is usually limited because of software and time constraints.

Reacting to these difficulties, the industry is using IR 4.0 tools more and more. Artificial intelligence, big data analysis, machine learning and automated control systems are helping engineers rethink the established processes. This change shows up in more intelligent, responsive, data driven systems that constantly adapt to the field conditions and operational feedback in gas lift design.

A digitally integrated gas lift design is no more only a theoretical idea. Fields around the world are already trialing real-world implementations such as intelligent gas lift valves, real-time SCADA monitoring and automated lift gas control (Al Qahtani et al, 2021; Dwivedi et al, 2022). Many operators still depend on strict, Excel-based processes with little integration or validation across platforms despite these developments.

This work presents a strong, modular, completely automated gas lift design framework developed in-house using hybrid approach combining the computational strength of petroleum engineering simulators with the adaptability of automation and integrated data mining tools. Over a broad range of well conditions, the system generates high fidelity gas lift designs, evaluates dynamic valve behavior, captures production uncertainty and imposes operations constraints.

The architecture, simulation logic, application to the real field data, performance benchmarks and future enhancement prospects of the system will be presented in the rest of this work. Not only the flow modelling but also the lifecycle adaptability, temperature pressure-correction mechanisms, valve envelope generation and the ideal mandrel depth selection is part of the design approach.

Embedding smart analytics and automation into the gas lift design process helps the system to fulfil both technical excellence and operational practicality. It provides a road map for future of gas lift design in digital oil fields, where real-time data and adaptive automation augment engineering intelligence.

Background and Gas Lift Fundamentals

From its first application in the early 20th century, gas lift is now one of the most commonly used forms in the artificially lifted wells (Oladoyin et al. 2020). In offshore, deviated and high GOR wells, its simplicity and robustness in demanding conditions and ability to lift fluids without moving downhole components make it a preferred choice. Fundamentally, gas lift lowers the hydrostatic head in production tubing by injecting high pressure gas into the produced fluid column, thereby lowering the FBHP and increasing inflow from the reservoir (Decker 2008).

Principles of Gas Lift Operation

Gas lift systems run under a dual flow concept. Between the tubing and casing, high pressure injection gas travels through the TCA and passes via gas lift valves (GLVs) into production tubing. The gas bubbles through the fluids column once inside the tubing, lowering the mixture density and allowing greater production rates. The flow behavior can be categorized into:

1. **Unloading Phase;** Gas is injected through upper valves to displace completion fluids and initiate flow.
2. **Transition Phase;** As tubing pressure decreases, deeper valves start to allow gas injection sequentially.
3. **Steady-State Production;** Gas is injected through the deepest active valve to maintain continuous flow.

The GLVs are set in side-pocket mandrels at the pre-defined depths and are activated at a pre-determined pressure. To regulate when and how much gas is injected, every valve is built with a specific port size, dome pressure with an R-Value (ratio of port and bellows areas).

Usually, a gas lift design calls for figuring the ideal valve spacing and opening-closing pressures for every valve. This guarantees lowest gas wastage, maximum drawdown and effective unloading. The complexity results from changes in the reservoir conditions brought about by depletion, water breakthrough or GOR variations. Dynamic, adaptive designs are necessary for the gas lift system to be flexible enough to react as these variations influence inflow and fluid gradients.

Types Of Gas Lift Valves

There are three main types of GLV used in modern designs:

1. **Injection Pressure Operated (IPO) Valves;** These are the most common, designed to open when casing pressure exceeds the dome pressure by a specific margin.
2. **Production Pressure Operated (PPO) Valves;** These are less common and are designed to open and close based on the tubing pressure.
IPO and PPO are commonly referred to as the unloading valves as they are only used during the unloading sequence during well startup.
3. **Orifice valve;** These are simple fixed orifice devices that allow continuous injection. They do not have dome or bellows and as such cannot open and close like the IPO/PPO valves. They are typically used as operating valves after unloading process is complete.

Advanced gas lift systems may also employ pilot-operated valves or smart valves with telemetry and adjustable porting, though these are not yet common due to cost and reliability concerns.

Key Design Parameters

The success of gas lift system hinges on the accurate estimation and integration of multiple subsurface and operational parameters. These include:

- **Reservoir Pressure (P_r):** Determines the drawdown available. As reservoir pressure depletes, deeper valve activation becomes necessary.
- **Productivity Index (PI):** Defines the relationship between reservoir pressure and flowrate.
- **Gas Liquid Ratio:** Affects both lifting efficiency and separator capacity. Higher GLRs reduce fluid density but may overload surface handling systems.
- **Kill or Completion Fluid Gradient:** Important during unloading phase when valves are initially opened against heavy fluid.
- **Tubing-Casing Gradients:** Influence pressure traverse calculation and determine the casing-tubing pressure differential, that drives valve actuation.
- **Well Deviation Profile:** Impacts valve installation and retrievability.
- **Valve Type & Performance Curve:** The valve selection, its size, type as well as dome pressures and port size directly impact injection sequencing.

- **Lift Gas Properties:** Includes gas gravity, compressibility factor and temperature. These factors determine the density and energy of the injected gas.
- **Lift Gas Injection Rate Availability:** Key component that ensures that the gas lift design is realistic in context of available gas supply and compressor capacity.
- **Corrosion and Erosion Considerations:** High velocity gas injection or wet sour gas environments require material compatibility assessments and appropriate port sizing.

An effective design must balance all these variables to provide a safe, efficient and robust gas lift system across all expected operating conditions.

Lifecycle Considerations in Design

Accounting for how reservoir and production system behavior changes over time is among the most underappreciated difficulties in gas lift design. A design that performs well in the first production phase could turn out to be less than ideal or perhaps non-functional later. Water breakthrough, for instance, can greatly change the fluid gradients, scale or sand production can affect pressure losses or valve performance.

Engineers used overdesigning for late-life scenarios or conservative assumptions historically to address these issues. This resulted in early production inefficiency and high gas lift injection rates. Modern design philosophy supports lifecycle-based optimization whereby valve spacing, injection depths and port sizes are chosen to enable reconfiguration or activation of various mandrels depending on the real-time feedback. Engineers can replicate scenarios reflecting reservoir decline, changing PI, changing GOR and increasing water-cut using forecasted profiles for early, mid and late-life conditions. This proactive approach guarantees that the gas lift system keeps running as best it can without calling for complete workovers as might be the case with some of the other artificial lift types.

Pressure-Temperature Considerations

Modelling of the pressure and temperature gradients across the wellbore is fundamental to generate a technically robust gas lift design. Not only mechanical construction but also the pressures and temperatures downhole, which vary greatly during unloading and production, control the opening and closing behavior of GLVs.

Valve Actuation and Dome Pressure Behavior

Most modern GLVs are IPO with nitrogen filled dome chamber that control the opening pressure. The valve opens when the casing pressure at the valve depth exceeds the dome pressure by a predetermined differential (often called cracking pressure). This differential is not static and is influenced by:

- Changes in casing pressure due to surface choke control or gas availability.
- Fluctuations in tubing pressure based on inflow, gradient and separator conditions.
- Variation in dome pressure due to thermal expansion of nitrogen gas.

As nitrogen is a compressible gas, the pressure inside the dome must be adjusted based on the downhole temperature to avoid premature or delayed opening. The industry standard practice is to calculate dome pressure at test rack temperature and then apply a correction factor for downhole conditions ([Sutton 2014](#)).

Effects Of Temperature on Gas Properties

In both annulus and tubing, temperature affects not only dome pressure but also the gas density and viscosity, thereby affecting flow dynamics and pressure losses. Particularly in prolonged unloading, cooler injection

gas may absorb heat from surrounding formations, causing contraction and pressure drop leading to unstable behavior. In high-moist environments, temperature drop can also lead to hydrate formation. When creating a new digital twin of the well, accurate temperature modelling is therefore crucial. Static geothermal gradients by themselves are not enough. Engineers have to replicate temperature profiles under flowing conditions, consider heat exchange between fluids, wellbore steel and formation rocks. Modern specialized simulators let multi-layer thermal models that can be used to enhance the realism of valve performance simulations.

Dome Pressure Adjustment

Precise correction of dome pressure from test rack conditions to downhole conditions is a fundamental component of gas lift valve design. Usually in bench testing or shop calibration, the dome of gas lift valve is pre-charged with nitrogen gas at surface temperature. Once downhole the valve is subjected to high temperatures and pressures, which changes the pressure of nitrogen charge due to thermal expansion and compressibility effects. If not adequately considered, this can greatly affect the opening pressure of the valve, leading to either early valve actuation or failure to open within the desired depth range.

Should this correction not been made, the valve would remain closed under expected opening casing pressures, causing delay in unloading or non-activation of the running valve. This shows the need of including thermal correction computations into every gas lift completion. Generating depth-specific temperature profiles from digital twin and dynamically computing dome pressures to match operational conditions, the in-house tool applies it automatically for every IPO valve. This results in creating the gas lift design that has a high degree of reliability and operational predictability.

Pressure Traverse and Temperature Calculations

Any gas lift design starts with accurate pressure traverse calculations since they control the annular and tubing pressures at every valve depth. These pressures control the final operating point stability, unloading sequence and valve opening order. Usually computed with nodal analysis combining frictional, hydrostatic and acceleration pressure losses, the traverse is calculated for tubing as well as TCA, as follows:

- **Tubing Pressure Traverse;** Includes the pressure drop due to multiphase fluid (oil, gas, water) moving from reservoir to surface. This accounts for:
 - Pressure losses due to friction
 - Hydrostatic head based on fluid density
 - Acceleration terms due to velocity increase
- **Casing Pressure Traverse;** Represents the pressure of injection gas descending through the annulus. Key considerations include the follow:
 - Gas compressibility
 - Pressure losses due to friction (minor in most annular flows)
 - Cooling induced contraction

Best Design Practices

- Use temperature corrected dome pressures at each valve depth
- Avoid placing GLVs in highly deviated sections due to diminishing returns and access issues (maximum deviation varies between companies)

- Validate pressure traverse using flowing bottom hole gauges or pressure-temperature surveys where available
- Account for separator or pipeline pressure fluctuations at surface, which impact WHP and affect valve sequencing
- Incorporate transient simulations for unloading stages with heavy kill fluid to assess the worst-case scenarios

Development of The In-House Gas Lift Design Framework

The in-house gas lift design framework has been developed to automate the entire design process in line with digital oilfield operations. The tool evaluates multiple scenarios, adapt to changing well conditions and guarantee repeatability across wells. Traditional process is often built around static excel sheets and manual interpretation of nodal analysis results for design purposes are progressively inadequate. This framework's aim was to include sophisticated engineering logic, data mining capability and scenario-based automation into single platform that smoothly interacts with industry leading production optimization applications.

The tool's framework can consume and organize vast amounts of data coming from several sources. To extract the important data including well schematics, deviation surveys, tubing-casing configurations, PVT properties, reservoir performance forecasts, historical production data and lift gas specifications, the tool interacts with internal well databases. Algorithm based validation system ensures completeness, unit consistency and logical integrity once this data is gathered. The tool then performs dynamic coupling with the well digital twin using validated data, writing input files for simulations as control engine, enabling a complete suite of sensitivity analysis. It produces and runs over 100 scenarios covering early, mid and late life conditions for a normal well. Key inputs for every simulation change include variable such as reservoir pressure, PI, GOR, FWHP, available gas injection pressures/rates and completion fluid gradient.

The embedded logic guides the valve depth selection and spacing while honoring physical operational constraints such as maximum deviation angles for slickline retrievability, maximum allowable injection pressures and minimum port size threshold. Computed for each valve, the valve design module generates critical specifications including test rack opening pressure, dome charge pressures (corrected for downhole temperature), R-values and port sizes.

Generation of output marks the last phase of the process. Along with the gas lift performance envelopes that plot liquid production rates against varying gas lift injection rates for various well life stages, the tool generates valve design sheets summarizing specifications for every mandrel. A design audit trail covering every assumption, simulation outcome and computations applied in the design balance is made available with the visual outputs. Along with providing transparency, this helps with peer review and troubleshooting while identifying future optimization opportunities.

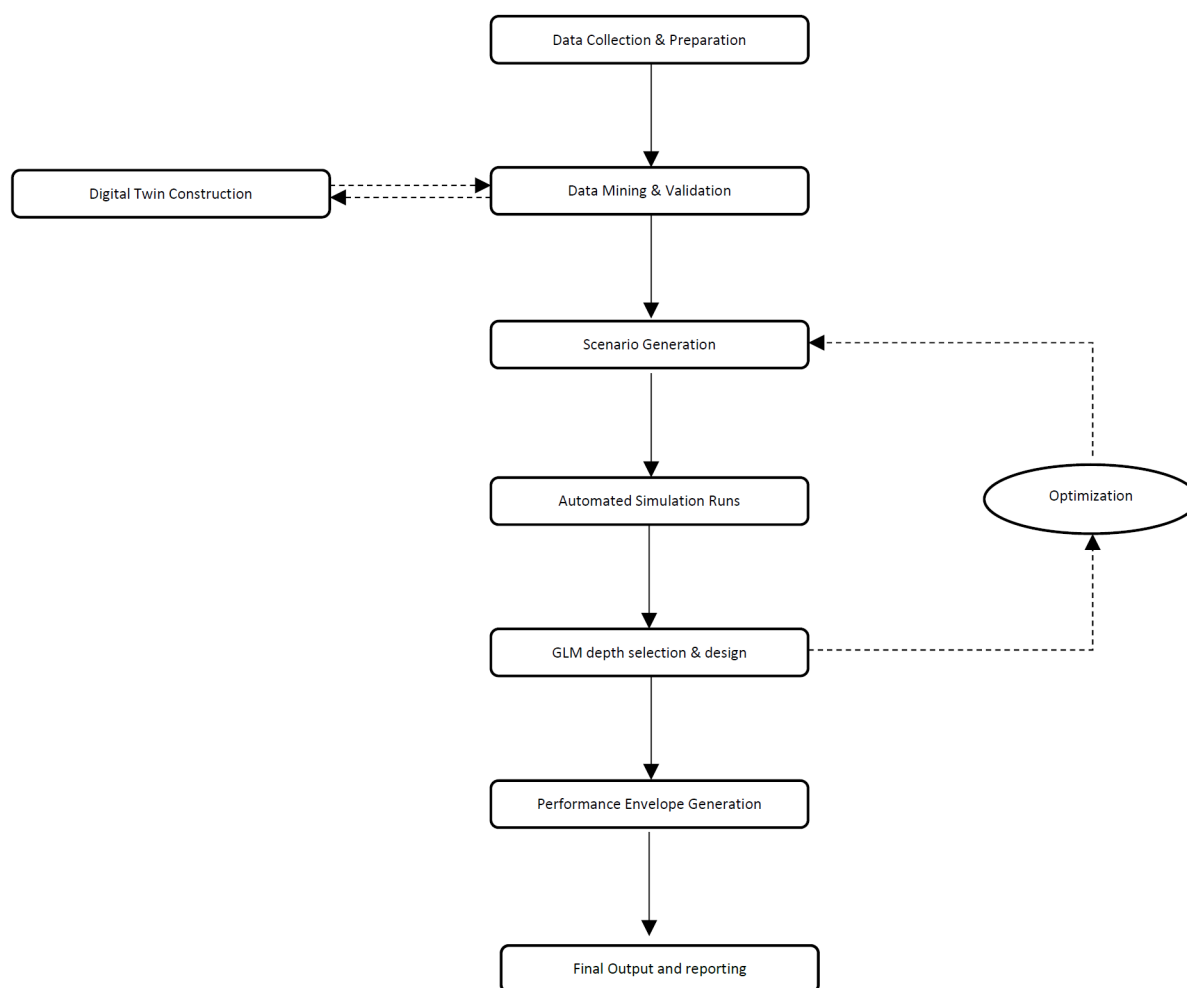


Figure 1—Gas Lift Design Tool Workflow

Case Applications and Simulation Results

A thorough study on a representative Well was undertaken to validate the design and field applicability of the in-house design tool.

The design tool was used to create a fit-for-purpose lifecycle gas lift design that could support effective unloading, minimize intervention work and sustain production over changing reservoir conditions. Starting with the digital twin of the well created with validated inputs including reservoir inflow parameters, wellbore geometry, surface facility limitations and gas lift availability. Production forecasts were incorporated to constraint to generate the optimum design.

The tool ran hundreds of simulations automatically covering a broad spectrum of gas injection scenarios, flowing pressures, GOR's and reservoir pressures. these simulations found a range of valve depths that offered best unloading flexibility and operational continuity. The expected function of valves during unloading, pressure transfer during intermediate staging or continuous operations was extensively modelled with specific design details. Additionally, strictly enforced design limitations including restricting the placement to within reasonable deviation angles for retrieval.

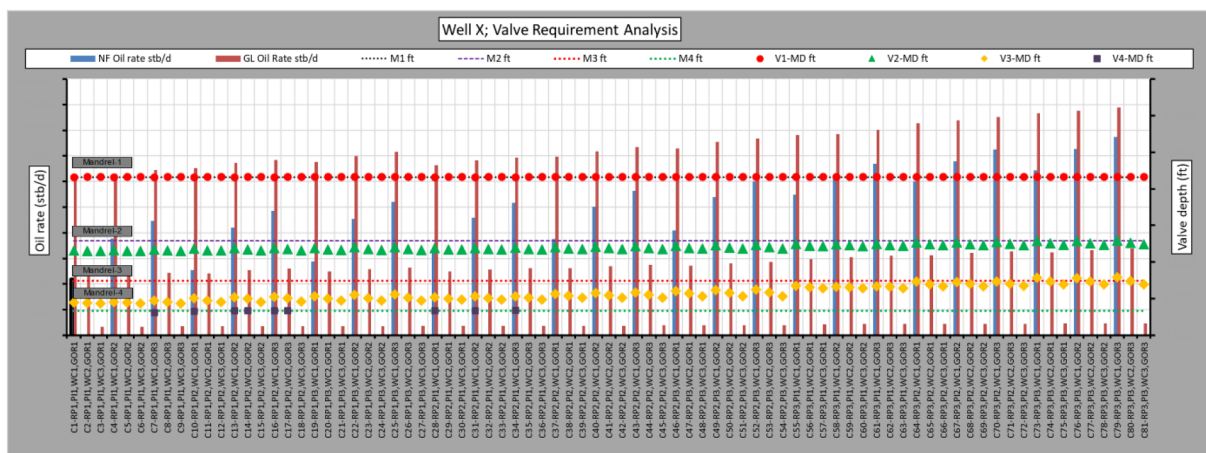


Figure 2—Valve requirement over well life

Every gas lift mandrel also called as station is assigned specific valve design including test-rack opening pressure (TRO), dome pressure corrected for downhole temperature and suitable port size to balance injection efficiency with flow stability. Based on thorough pressure and temperature profiles obtained from the dynamically coupled digital twin with the design tool.

Gas lift performance envelopes were generated for every phase along the well life. These envelopes showed the link between gas injection rate and liquid production, enabling engineers to choose the most effective injection technique for every condition. Concurrent with the design, a theoretical analysis was also performed to understand whether a deeper valve beyond the standard retrievable depth limit would produce significant production uplift. Although the late life situations showed some improvement, the higher operational risk linked with deeper valve placement exceeded the advantage. Therefore, the tool advised preserving the deeper valve option as a backup while maintaining the valve depth within the retrievable limits.

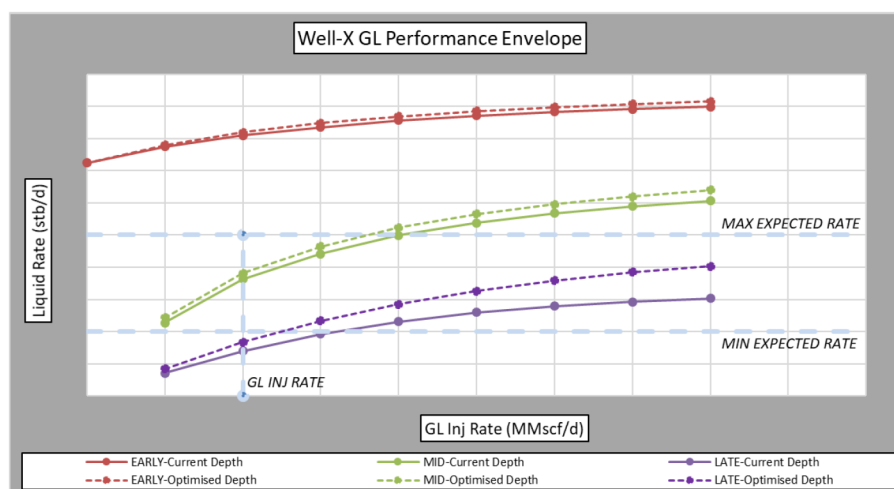


Figure 3—Gas Lift performance envelope

A design report was produced that combined all simulation results, valve designs and envelope plots. The peer reviewed and approved design was shared with the completion and work-over teams for valve procurement, mandrel placement with a view of ultimate deployment. Additionally, all the design files were also made available for performance benchmarking and future use. This study underlined the importance of

tool's automation, consistency and capacity to generate bespoke fit-for-purpose designs for changing well conditions.

Areas of Improvement & Future Enhancement

There are several opportunities for improvement even if the gas lift design tool marks a major breakthrough in automation, lifecycle adaptability and its capability to integrate with the industry standard petroleum engineering applications. The improvements line up with the larger goal of completely autonomous artificial lift system and real-time operational control including gas lift design. Integration of live SCADA data streams into the logic of the tool is one main area for upcoming improvement. Establishing a link between the simulation engine and field data would enable the tool to go from a design only platform to real-time monitoring and optimization system. Should the tool find a difference between expected and tested flowing parameters, for instance, it could advise changes in injection rate, notify field engineers about valve issue such as shallow lifting or flag conditions likely to cause multi-point injection.

Integration of machine learning techniques to predict ideal valve depths and injection strategies depending on the past or offset well performance across a field or asset would also be another highly impactful improvement. Training the system on previously successful lift techniques would allow the design engine to not only replicate but also suggest the most effective layouts based on analogue wells. This would be especially helpful in a field with vast amount of gas lift wells where consistency and speed are critical.

From a user interface standpoint, 3D visualization features could help future tools to show mandrel positions, deviation angles and valve envelopes in a spatial environment. This would improve cross-functional communication and give a clearer knowledge of valve accessibility, deviation related hazards and ease of retrieval, particularly with operations and completion.

Direct export of the design results straight into execution processes such as slick-line run sheets, rig less and pre-filled workover programs. This would reduce the intervals between design and implementation and would greatly increase field readiness and response times.

Ultimately, the system will be upgraded to incorporate automatic compliance checks for criteria including API 19G2 qualification, H_2S service rating and operational envelope reviews as regulatory criteria and internal governance standards evolve. These checks will be included into design logic so lowering the possibility of non-confirming hardware finding their way into the field.

The improvements taken together will convert the tool from design accelerator into fully functional lift management system. It will aid in optimizing individual well output while lowering the compressor loading and fuel gas requirements.

Conclusion

An important turning point in the progress of artificial lift engineering is the development and implementation of a digitally integrated, IR 4.0 enabled gas lift design tool. This method presents engineers with a robust substitute for conventional spreadsheet-based processes by including scenario-based modelling, dynamic pressure-temperature corrections and embedding automation into a centralized system. It redefines not only the generation but also the communication, review and application across operational teams of gas lift designs. The presented study shows how the system can convert difficult subsurface and surface constraints into useful, field ready design that maximizes production across the lifetime of the well. The tool generates technically sound, logically reasonable and operationally strong gas lift configurations by simulating hundreds of scenarios in minutes, automatically applying dome pressure correction and enforcing critical constraints including maximum wellbore deviation.

Moreover, the utility of the tool goes beyond designing stage. The design output forms the common language for cooperation among subsurface engineers, completion teams, procurements departments and field operations, such as mandrel details and depths selection, valve specification, performance envelopes

and pressure traverses. Particularly in big field development projects, this common knowledge lowers execution risks and encourage consistence across wells. Tools like this are not optional as the industry negotiates operational complexity, cost constraints and the drive for automation. It prepares the engineering teams to transition from reactive troubleshooting to proactive, statistically based design and monitoring. Intelligent systems that fit the reservoir, learn from the field and always improve themselves will have artificial lift especially gas lift into the future. This paper shows not only the viability of such systems but also their instant worth in upstream activities of today. This method will significantly help production sustainability, cost efficiency and digital transformation in oil and gas sector as it is further polished and scaled.

Nomenclature

AI	Artificial Intelligence
API	American Petroleum Institute
ESP	Electric Submersible Pump
FBHP	Flowing Bottomhole Pressure
GOR	Gas Oil Ratio
GLV	Gas Lift Valve
H_2S	Hydrogen Sulfide
IR 4.0	Industrial Revolution 4.0
IPO	Injection Pressure Operated (valve)
PVT	Pressure Volume Temperature
SCADA	Supervisory Control and Data Acquisition
TCA	Tubing Casing Annulus
TRO	Test-rack Opening Pressure
WHP	Wellhead Pressure

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